

The Coil Dividend: A Global Health-Economics Study of Neurovascular Head Coils for Tumor Therapy

Economist-style briefing article

Executive summary

Neurovascular head coils, once treated as a narrow imaging input, are increasingly becoming a strategic therapeutic platform in brain-tumor care. Their economic value no longer depends only on scanner-room throughput; it also depends on how much they change the full treatment pathway: time-to-diagnosis, targeting precision, repeat intervention rates, edema management, hospitalization days, and long-horizon survival-adjusted quality of life.

This study develops a global framework that combines three uncommon but useful tools:

1. Continued fractions for stable, multi-stage value attribution under uncertainty.
2. Combinatorial game theory for real-options pricing across payer-provider-manufacturer strategy sets.
3. Inflection-point distributional modeling for nonlinear adoption and outcomes dynamics across countries.

Across baseline simulations, value is highest when coils are deployed with protocol standardization and reimbursement innovation, not as standalone hardware upgrades. Middle-income systems show the largest marginal welfare gains when financing is staged and linked to measured quality-adjusted life-year (QALY) conversion.

The 2060 outlook adds a second dividend: decarbonized medical-device pathways. Green redesign of coils, stents, catheters, and implants can reduce lifecycle emissions and hazardous waste while preserving clinical performance, making sustainability a measurable component of net social value.

1. Why head coils matter economically in tumor therapy

In neuro-oncology, delayed or noisy signal acquisition propagates expensive errors downstream. Better neurovascular head coils can improve signal-to-noise ratio, vascular characterization, and lesion boundary confidence, especially in treatment monitoring and thermometry-linked interventions. The direct cost delta of coil acquisition is visible; the larger economic effect sits in avoided false negatives, fewer non-therapeutic procedures, and improved treatment matching.

Global heterogeneity is the central policy problem:

- High-income systems face diminishing returns unless advanced coils are integrated into adaptive clinical pathways.
- Upper-middle-income systems face financing and procurement frictions despite large potential gains.
- Lower-resource systems face workforce and maintenance bottlenecks that can mute technical benefits.

The policy question is therefore not "Should systems buy better coils?" but "Under what institutional game do coils become welfare-accretive at scale?"

2. Continued-fraction value architecture

Conventional cost-effectiveness often collapses pathway dynamics into a flat incremental cost-effectiveness ratio. We instead model recursive value conversion:

$$[Equation: V = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \ddots}}}]$$

where each a_i is a stage friction term:

- a_0 : immediate diagnostic yield gain
- a_1 : workflow integration friction
- a_2 : reimbursement delay friction
- a_3 : learning-curve friction
- a_4 : maintenance and downtime friction

Interpretation: if any denominator term worsens sharply (for example, reimbursement lag), total realized value contracts nonlinearly. This reflects field reality more accurately than additive scoring.

For country c , hospital class h , and year t :

$$[Equation: V_{c,h,t} = \Delta QALY_{c,h,t} \cdot WTP_c - \Delta Cost_{c,h,t} \dots]$$

and the realized conversion efficiency is represented as a continued fraction multiplier $\Phi_{c,h,t}$:

$$[Equation: V^*_{c,h,t} = V_{c,h,t} \cdot \Phi_{c,h,t} \dots]$$

with $0 < \Phi_{c,h,t} < 1$ in most deployments. This creates a tractable bridge between engineering performance and institutional execution risk.

3. Combinatorial game theory and options pricing

3.1 Strategic players

We define a finite game among:

- Public or social payer (P)
- Provider network or hospital chain (H)
- Coil manufacturer and service partner (M)

Each has discrete strategy sets:

- P: fixed tariff, outcome-linked tariff, staged subsidy
- H: defer adoption, pilot adoption, full rollout
- M: capital sale, managed service, risk-sharing contract

The strategy profile space is combinatorial: $|S| = |S_P| \times |S_H| \times |S_M|$. The payoff for each player depends on adoption timing, measured outcomes, and demand uncertainty.

3.2 Real option valuation

Treat adoption as a compound real option. Pilot deployment grants the right, not obligation, to scale. Option value rises with clinical-outcome volatility when downside is contractually shared.

A simplified option payoff for hospital h is:

$$[Equation: \Pi_h = \max\left(0, \sum_{t=1}^T \frac{(R_t - C_t) \cdot \theta_t}{(1+r)^t} - I_0\right) \dots]$$

where θ_t is state-dependent clinical conversion efficiency linked to coil-enabled outcome gains.

Combinatorial game equilibrium identifies contract structures where:

- P limits budget risk
- H retains upside from reduced downstream burden
- M internalizes part of performance risk

In baseline runs, equilibrium most often selects staged subsidy + pilot adoption + risk-sharing service contracts, especially in upper-middle-income regions.

4. Inflection-point distributional relationships

Adoption and outcomes are not linear. We model two linked S-curves:

1. Technology adoption share
2. QALY-to-budget conversion efficiency

For adoption share $A(t)$:

[Equation: $A(t) = \frac{1}{1 + e^{-k(t-t_0)}} \dots$]

The inflection point at $t=t_0$ determines when marginal diffusion is fastest.

For distributional outcomes, let net benefit density be $f(x)$ with a policy-relevant curvature threshold where:

[Equation: $\frac{d^2 f}{dx^2} = 0 \dots$]

This inflection identifies where incremental subsidy shifts from high-yield to low-yield spending.

Empirically, three patterns recur globally:

- Before inflection: workforce and protocol training dominate hardware quality.
- Around inflection: reimbursement design dominates speed of value realization.
- After inflection: marginal gains depend on interoperability, preventive pathways, and data quality.

This distributional lens prevents overinvestment in late-stage markets while underfunding takeoff markets.

5. Global scenario results (stylized)

Scenario A: Status quo procurement

- Upfront hardware purchases with weak service guarantees.
- Slow pathway integration.
- Estimated welfare gain: low to moderate; high variance.

Scenario B: Outcome-linked procurement

- Payment tied to predefined diagnostic and treatment-monitoring milestones.
- Shared downside risk between provider and manufacturer.
- Estimated welfare gain: moderate to high; variance reduced.

Scenario C: Staged global financing with local capability build

- Early grant or blended finance for pilot sites.
- Trigger-based expansion after crossing inflection metrics.
- Embedded workforce upskilling and uptime commitments.
- Estimated welfare gain: highest in middle-income systems; strongest equity-adjusted returns.

Across scenarios, sensitivity analysis shows the strongest drivers of net value are:

- Reduction in repeat intervention probability
- Time-to-therapy acceleration
- Maintenance uptime and protocol adherence
- Payment architecture (especially lag-to-reimbursement)

6. Equity and macroeconomic spillovers

Better coil-enabled pathways can reduce catastrophic health spending through earlier and more accurate treatment planning, especially where households finance care out-of-pocket. Macro spillovers include lower productivity loss from prolonged disability and reduced pressure on tertiary referral centers.

A practical equity metric is distribution-weighted net monetary benefit:

$$[Equation: DW\text{NMB} = \sum_i w_i (\Delta QALY_i \cdot WTP_i - \Delta Cost_i) \dots]$$

where w_i upweights vulnerable populations. Under distribution weighting, staged adoption in underserved urban-peripheral hubs frequently outranks immediate saturation in already advanced centers.

7. Green-device transition: eco-friendly footprint by design

The "Xerox cartridge printer" metaphor is useful: hospitals should not buy only a device shell; they should procure the full consumables-service-recycling ecosystem. In imaging and intervention, this means contracting not just head coils or delivery hardware, but the lifecycle package around sterilization, replacement parts, biocompatible materials, reverse logistics, and end-of-life recovery.

7.1 Device-level decarbonization targets

- Neurovascular head coils: increase recycled copper and low-VOC composites; modularize electronics for repair over replacement.
- Stents: prioritize low-energy manufacturing routes and recyclable packaging; reduce revision rates through better matching and guidance.
- Catheters: shift toward lower-toxicity polymers, reduced single-use plastic mass, and sterile reprocessing where clinically valid.
- Implants: reward longer durability and lower revision burden, as repeat surgeries carry both clinical and carbon costs.

7.2 Sustainability-adjusted net monetary benefit

Extend valuation with carbon and waste terms:

[Equation: $SA_{NMB} = \Delta QALY \cdot WTP - \Delta Cost - \lambda_C \Delta CO_2e - \lambda_W \Delta Waste \dots$]

where λ_C is the social cost of carbon and λ_W is the shadow price of regulated medical waste. Positive SA_{NMB} identifies strategies that are both clinically and environmentally accretive.

7.3 Relational procurement contracts

Under the relational paradigm, each party is paid for system performance, not item volume:

- Payers reimburse outcome bands plus verified emissions-intensity reduction.
- Providers receive gain-sharing for reduced complications, shorter stay, and waste minimization.
- Manufacturers receive bonus payments for repairability, take-back rates, and material circularity.

This converts the historical "sell-more-units" equilibrium into a "deliver-more-health-per-tonne" equilibrium.

8. Policy blueprint for a global rollout

1. Use continued-fraction diagnostics to map where value is being lost along the pathway.
2. Procure through combinatorial contract design, not one-shot hardware tenders.
3. Price adoption as a real option: pilot, learn, scale.
4. Track inflection indicators quarterly:
 - protocol adherence
 - reporting turnaround
 - repeat procedure rate
 - progression-free survival proxies
5. Tie expansion capital to verified crossing of inflection thresholds.

6. Add green key performance indicators (KPIs):

- lifecycle CO2e per treated patient
- regulated waste mass per treatment cycle
- repairability index and take-back rate
- share of recyclable or recycled material input

7. Use green public procurement rules by 2035 and harmonized disclosure by 2040 to avoid fragmented standards.

9. Economist-style conclusion

The world does not have a head-coil technology problem as much as a coordination problem. Neurovascular head coils for tumor therapy can be cost-effective, even cost-saving, but only when engineering upgrades are wrapped in incentive-compatible contracts and timed to diffusion inflection points.

In high-income systems, the prize is efficiency and precision. In middle-income systems, the prize is leapfrogging. In low-resource settings, the prize is avoiding a two-tier neuro-oncology future.

By 2060, the best-performing systems will treat clinical precision and environmental efficiency as one production function. The central economic insight is simple: value compounds when uncertainty is managed as an option, institutions are treated as strategic players, nonlinear diffusion is measured rather than assumed, and device lifecycles are priced for circularity.

Technical appendix (compact)

A. Continued-fraction multiplier construction

Define stage friction vector $\mathbf{a}=(a_1,\dots,a_n)$ and:

$$[\text{Equation: } \Phi(\mathbf{a}) = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\cdots + \frac{1}{a_n}}}} \dots]$$

Lower a_i from policy reform or training raises Φ superlinearly when the improved stage is near the top of the recursion.

B. Combinatorial game payoff matrix

For strategy profile $s \in S$:

[Equation: $U_P(s), U_H(s), U_M(s)...$]

Select Nash-stable profiles under budget and equity constraints. Real option value is embedded via state-contingent continuation payoffs.

C. Inflection-aware subsidy rule

Let $g(z)$ be marginal social return to subsidy intensity z . Use:

[Equation: $z^* : \frac{d^2 g}{dz^2} = 0...$]

to identify the pivot where subsidy should shift from acceleration to consolidation.